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ARGUS

Autonomous Real-time Gateway Unsupervised Shield

AI-POWERED REAL-TIME NETWORK SECURITY

In today's connected world, security is not an option — it's a necessity. Let's safeguard the digital age together.



OUR TEAM :

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THE GROWING DIGITAL SECURITY GAP:



As internet connectivity expands across homes, schools, hospitals, and rural communities, cybersecurity protection has not grown at the same pace.

Challenges:

- Weak router security
- No active monitoring
- Limited cybersecurity awareness
- Lack of dedicated security personnel

Consequences:

- Data theft
- Privacy breaches
- Financial loss
- Service disruption



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THE PROTECTION GAP:

Traditional Firewalls

- Static rules
- Limited visibility
- Misbehavioral anomalies

Enterprise IDS/IPS

- Powerful
- Expensive
- Complex deployment requiring experts

Small organizations need enterprise-level protection
without enterprise-level complexity.



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INTRODUCING ARGUS:

ARGUS is an AI-powered Network Anomaly Detection and Blocking System designed to provide intelligent cybersecurity protection for local networks.

Capabilities:

- Monitor network traffic
- Detect suspicious activity
- Classify threats
- Alert administrators
- Block confirmed attackers
- Visualize incidents in real time

Mission:

Making advanced cybersecurity accessible to every connected community.



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VISUAL FLOW:

Network Traffic → Traffic Analysis → Feature Extraction → Threat Detection →
Classification → Blocking → Dashboard

ARGUS continuously observes network behavior and identifies threats before they escalate.

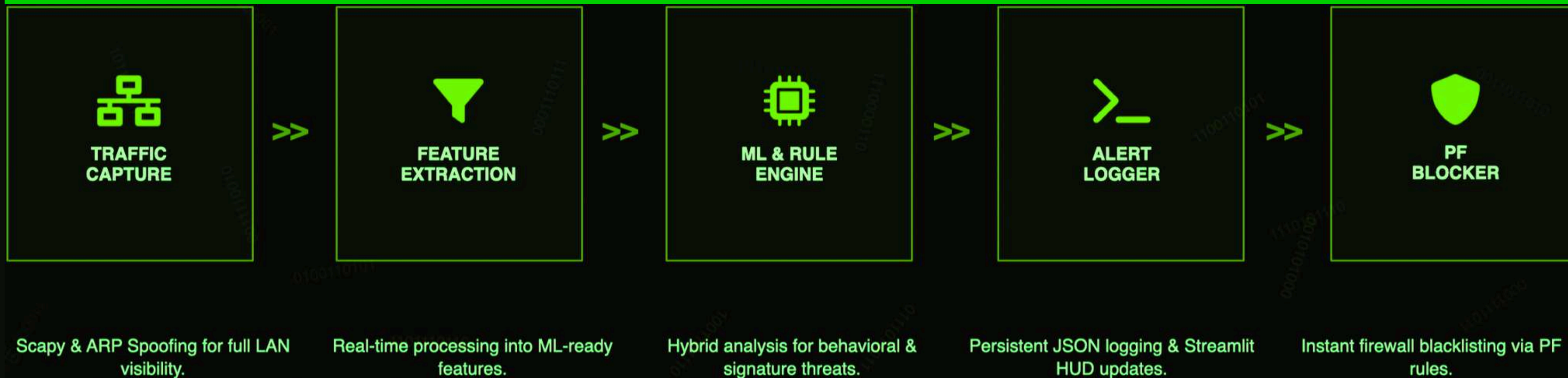


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COMPONENTS:

Traffic Visibility Layer, Packet Capture Engine, Feature Extraction Module, Machine Learning Engine, Rule-Based Detection Engine, Decision Layer, Blocking Engine, Dashboard Layer

Traffic is captured and analyzed in real time. Threat intelligence is generated using both ML and explicit signatures.





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HYBRID THREAT DETECTION:

Rule-Based Detection

Detects:

- Port scans
- Host sweeps
- SYN floods

Machine Learning Detection

Isolation Forest Model

Detects:

- Traffic anomalies
- Behavioral deviations
- Unknown attack patterns

Key Innovation:

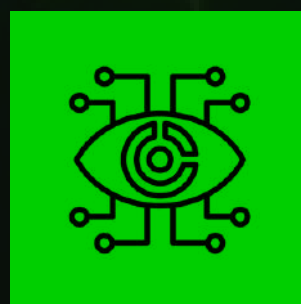
Known attacks are blocked immediately, while suspicious, unknown behavior is intelligently analyzed.





THREATS DETECTED INCLUDE:

- TCP Port Scans
- SYN Scans
- Connect Scans
- Host Sweeps
- Reconnaissance Activity
- SYN Flood Attacks
- Statistical Traffic Anomalies



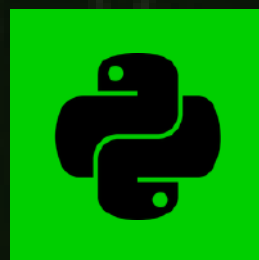


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TECHNICAL APPROACH:

Programming Language:

Python



Packet Analysis:

Scapy



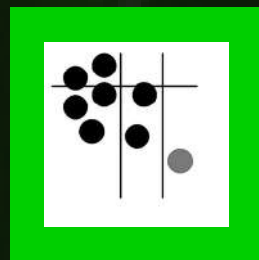
Machine Learning:

scikit-learn



Model:

Isolation Forest



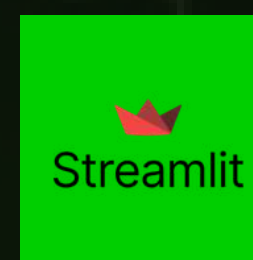
Data Processing:

Pandas, NumPy



Visualization:

Streamlit



Training Datasets:

- NSL-KDD
- KDDTest
- CICIDS2017



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REAL-TIME DASHBOARD:

Features:

- Live alert feed
- Threat severity scoring
- Attack classification
- Timeline analysis
- Source investigation
- Historical logs

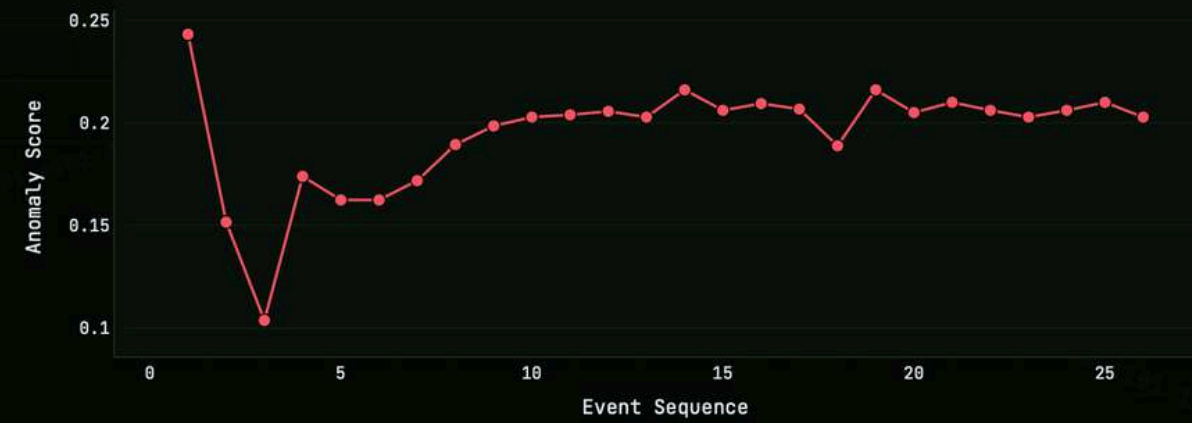




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REAL-TIME DASHBOARD:

ANOMALY TIMELINE



SOURCE PRESSURE



Source Profiles Port And Payload Evidence Raw Event Registry

EVIDENCE RANKING





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OPERATIONAL WORKFLOW:

Step 1:

Traffic Monitoring

Step 2:

Packet Capture

Step 3:

Feature Extraction

Step 4:

ML + Rule Evaluation

Step 5:

Threat Classification

Step 6:

Alert Generation

Step 7:

Blocking Decision

Step 8:

Dashboard Update

Attacker → Traffic → Detection → Classification → Alert → Block → Protected Network



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SAFEGUARDS:

- Trusted device whitelist
- Monitor mode
- Configurable thresholds
- Reduced false positives
- Safe shutdown
- Controlled automation

Important Statement:

Machine-learning anomalies alone are not automatically blocked without additional validation.





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IMPACT & BENEFICIARIES:

Beneficiaries:

- Homes
- Schools
- Hospitals
- Small Businesses
- Rural Digital Centers

Benefits:

- Improved cybersecurity access
- Reduced cybercrime risk
- Early attack detection
- Increased trust in digital systems





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SCALABILITY & FUTURE ROADMAP:

Phase 1:

Current ARGUS

Phase 2:

- Telegram Alerts
- Slack Alerts
- Threat Intelligence

Phase 3:

- Web Portal
- Mobile App
- Multi-Network Monitoring

Phase 4:

- Edge AI Security Appliance
- Federated Learning
- Enterprise Analytics





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PROJECT VISION:

Cybersecurity should not be a privilege reserved for large organizations.

ARGUS aims to democratize intelligent network protection and make advanced security accessible to every connected community.

FUTURE VISION:

Affordable AI-driven cybersecurity for everyone.





CONCLUSION:

ARGUS combines:

- Artificial Intelligence
- Behavioral Analysis
- Real-Time Detection
- Automated Protection

to provide practical
cybersecurity for underserved
networks.





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THANK YOU!

Thank you for being part of the secure digital journey.